

Year 8 - 2015



Baldivis
Secondary College

Investigation 2

Semester One Term 2

Circles

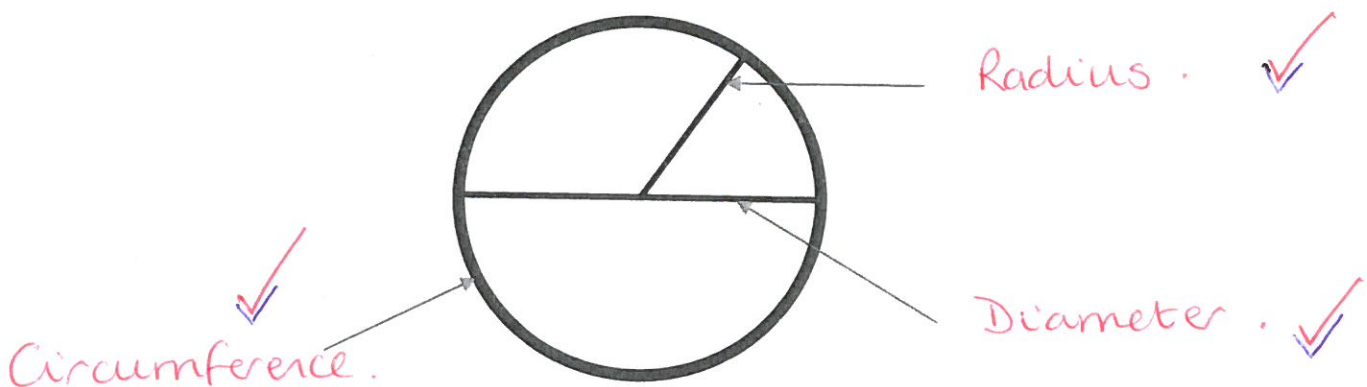
This investigation will be completed in class under test conditions. No guidance or hints will be provided. You will need access to a scientific calculator.

AIM: In this assessment you will be investigating the properties of circles and the relationship between circumference, area, radius and diameter. You will also use formulas to solve problems involving circumference and area. This will be done using both diagrams and calculations.

Name: Solutions . Class: _____ Total: _____ /20

1. Label the parts of the circle on the diagram

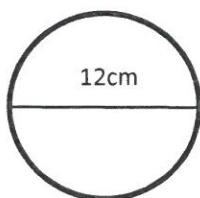
(3)



pie	radius	area	circumference	diameter
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2. Using the formula $C = \pi \times d$ find the circumference of the circle below

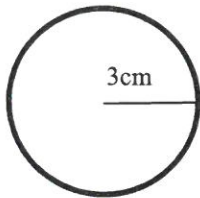
(1)



$$C = 12\pi \\ = 37.70 \text{ cm (2dp)}$$

(accept either form)

3. Circumference can also be found using the formula $C = 2 \times \pi \times r$. Use this formula to find the circumference of the circle below. Give your answer in exact form. (1)

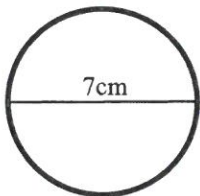


$$C = 2 \times \pi \times 3$$

$$= 6\pi \quad \checkmark$$

4. For the following circles, use the formulas you have already used to find their circumferences. Give your answer to 2 decimal place. (4)

a.

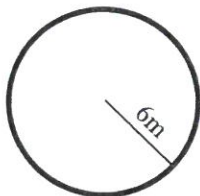


$$C = \pi \times 7$$

$$= 7\pi$$

$$= 21.99 \text{ cm} \quad (2 \text{ dp}) \quad \checkmark$$

b.

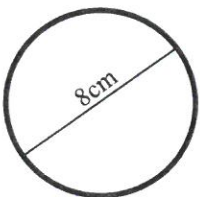


$$C = 2 \times \pi \times 6$$

$$= 12\pi$$

$$= 37.70 \text{ m} \quad (2 \text{ dp}) \quad \checkmark$$

c.

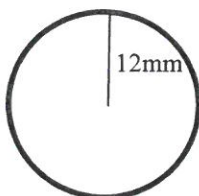


$$C = \pi \times 8$$

$$= 8\pi$$

$$= 25.13 \text{ cm} \quad (2 \text{ dp}) \quad \checkmark$$

d.

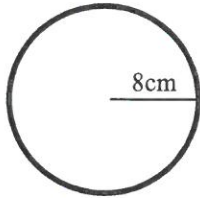


$$C = 2 \times \pi \times 12$$

$$= 24\pi$$

$$= 75.40 \text{ mm} \quad (2 \text{ dp}) \quad \checkmark$$

5. The area of a circle can be found using the formula $A = \pi \times r^2$. Use this formula to find the area of the following circle. Give your answer to two decimal places (1)



$$\begin{aligned} A &= \pi \times 8^2 \\ &= 64\pi \\ &= 201.06\text{cm}^2 \end{aligned}$$

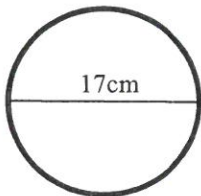


6. How would you use this formula to find the area of a circle if you only knew the diameter? (1)

Halve the diameter and use that value in the formula $A = \pi r^2$



7. Use your answer from question 6 to find the area of the circle below. Give your answer in exact form. (1)

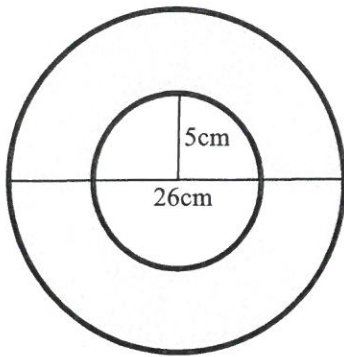


$$\begin{aligned} 17 \div 2 &= 8.5\text{cm} \\ A &= \pi \times 8.5^2 \\ &= 72.25\pi \end{aligned}$$



8. Use your knowledge of radius and diameter to find the width of the shaded section of this circle

(1)



$$26 \div 2 = 13$$

$$13 - 5 = 8 \text{ cm}$$



9. Use the measurements from the circle in question 8 to find

(4)

a. The area of the larger circle. Give your answer in exact form.

$$A = \pi \times 13^2$$
$$= 169\pi$$



b. The area of the smaller circle. Give your answer in exact form.

$$A = \pi \times 5^2$$
$$= 25\pi$$



c. The area of the shaded section. Give your answer in exact form.

$$169\pi - 25\pi = 144\pi$$

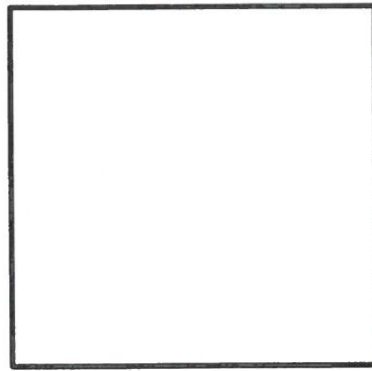


d. Now give your answer correct to 2 decimal places.

$$452.39 \text{ cm}^2$$



10. Ali has a square piece of paper that is 23cm in width. For each of the following questions, give your answer correct to 2 decimal places. (3)



- a. What is the circumference of the largest circle that it is possible to cut from this size of paper?
(*hint: draw and label a diagram using the square above as a guide)

$$d = 23\text{cm} \quad r = 11.5\text{cm}$$

~~Az~~

$$C = \pi \times 23$$

$$= 23\pi$$

$$= 72.26\text{cm}$$



- b. After the circle has been cut, what is the area of the paper left over?

$$\begin{aligned} \text{A of the square} &= 23 \times 23 \\ &= 529\text{cm}^2 \end{aligned}$$

$$\begin{aligned} \text{A. of the circle} &= \pi \times 11.5^2 \\ &= 132.25\pi \\ &= 415.48\text{cm}^2 \quad (2\text{dp}) \end{aligned}$$



$$\begin{aligned} \text{Area left over} &= 529 - 415.48 \\ &= 113.52\text{cm}^2 \end{aligned}$$



